

Ministry of Higher Education and Scientific Research
University of Algiers 1 - Faculty of Science
Mathematics Department

Module: Machine Structure 1
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Sector: Mathematics
Semester 1

Series of exercises No.2

Exercise 1:

Two numbers A and B represented in three positions as follows:

- $A=(a_3 a_2 a_1)_5$; $B=(b_3 b_2 b_1)_7$
 - 1- What are the possible values for the coefficients a_i , b_i ?
- We give: $A+B=(138)_9$, $A-B=(200)_6$.
 - 2- Find the values of the coefficients a_3 , a_2 , a_1 , b_3 , b_2 , b_1 .
 - 3- Make A and B in binary, then calculate A+B and A-B.

Exercise 2: Complete the following table

Decimal	Pure Binary	BCD	GRAY
226			
	11011101		
		0100100001100111	
			10011101

- In a digital system that manipulates 32-bit words, how many unsigned integers can be represented if they are encoded in:
a) Pure Binary and
b) BCD

Exercise 3:

- Give the number of values and the representation interval of unsigned integers on 4, 8, 16 and 32 bits.
- How many minimum bits are required to present the following numbers as unsigned integers?

(a) $(147)_{10}$; (b) $(384)_{10}$; (c) $(1560)_{10}$

Exercise 4:

- Give on 8 bits the representations “SVA”, “One’s Complement” and “Two’s Complement” of the numbers $(37)_{10}$ et $-(118)_{10}$.
- Decode the following 2’s Complement encoded signed integers on 8 bits:
(a) $(1100\ 1001)_{C2}$; (b) $(0110\ 1101)_{C2}$; (c) $(1000\ 000)_{C2}$.

Exercise 5:

- Give the decimal value of the binary numbers **(11111111)**, **(10010001)** each considered in :
 - An unsigned integer encoding on 8 bits,
 - Signed by **SVA**, or signed in **One's Complement** or signed in **Two's Complement**.

Exercise 6:

- Encode **(25)₁₀** and **(-25)₁₀** on one **byte (8 bits)** using **sign and absolute value (SVA)** representation.
- Add between **(25)₁₀** and **(-25)₁₀** in **sign and absolute value (SVA)**.
- Encode **(25)₁₀** and **(-25)₁₀** on **one byte** using **Two's complement (C2)** representation.
- Add between **(25)₁₀** and **(-25)₁₀** in **Two's complement**. What do you notice?

Exercise 7:

- Consider three numbers represented in 2's complement: **N1 = 00110101**; **N2 = 01101100** and **N3 = 11010111**. Perform the following arithmetic operations. Specify for each case whether there is an overflow.
 - a. $N1 + N2$ b. $N1 + N3$ c. $N2 - N1$

Exercise 8:

- Encode the following numbers in single-precision floating point IEEE754. Give the result in hexadecimal.
 - a) -1.5 ; (b) 1.5 (c) 11,625
- Find numbers equivalent to the following numbers encodes in IEEE754 :
 - (a) 11000010110101010100000000000000
 - (b) 3F 80 00 00

Exercise 9:

- Consider two numbers encoded according to the IEEE 754 standard and represented in hexadecimal:
3EE00000 and 3D800000
 - ✓ Calculate the sum and give the result in IEEE 754 form and decimal form.
 - ✓ Calculate the multiplication and give the result in IEEE 754 form and decimal form.

Exercise 10:

Consider the extract of the code written in a C programming language, which has the following types:

- **Unsigned** : 8-bits unsigned integers
- **Float**: Real numbers represented in IEEE754 single precision on 32 bits
- **Int**: signed integers represented in 2's Complement on 16 bits

1- Part of Numbering system :

- a) Calculate A + B on base 16. (do the operation in base 16)
- b) Calculate B - A on base 8.
- c) Represent B in Gray code.

2- Part of Signed integers :

- a) Represent the numbers C and D in C2.
- b) Calculate C+D in C2 and specify if there is overflow. Justify your answer

3- Part of real numbers :

- a) Calculate X+Y in floating point then give the notation of the result in Hexadecimal and decode it in decimal.

```
main () {  
    unsigned A,B;  
    int C,D;  
    float X,Y;  
    .....  
    .....  
    A = 36;  
    B = 128;  
    C= - 47; D= +92 ;  
    X = - 61.5;  
    Y = 0.125  
    .....  
}
```

Exercise 11:

A book has an average of 800 pages. If each page is made up of 90 columns and 50 rows. How many medium sized books can be stored on an 8 GB size USB memory?

Exercise 12:

- What does the following binary sequence represent if it is encoded in ASCII:

1010011101010010011111010000

- Which bit must be complemented to change an uppercase ASCII letter into a lowercase ASCII letter and vice versa?

Exercise 13:

- Either the sentence coded in ASCII by the following hexadecimal code sequence:

41 53 53 49 53 54 45 52 20 41 55 20 43 4F 55 52 53 20 45 54 20 54 44 20 45 53 54 20 49 4D 50 4F 52 54 41 4E 54

- 1- How many distinct characters does this phase contain?
- 2- Decode this sentence